



Exploratory Data Analysis using Python

Analyzing 30 Years of Global Metal Price Trends

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Python · Pandas · Matplotlib

Why metal prices matter

This project studies long-run price behavior for aluminium, gold, nickel, silver, and uranium to understand commodity cycles, inflation impact, and market volatility patterns across 1992–2021.

Dataset size

361 × 13

monthly observations and derived features

Time span

1992–2021

about 30 years of market history

Use case

**Investment +
manufacturing**

pricing, risk, and procurement insights

Source

Kaggle

commodity price trend dataset

METALS COVERED

Aluminium

Gold

Nickel

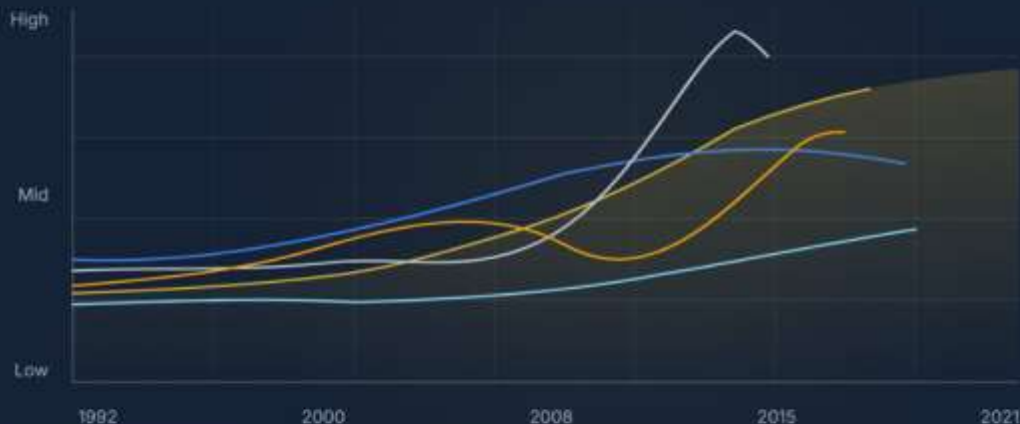
Silver

Uranium

Inflation-adjusted pricing improves interpretation because nominal gains can overstate true purchasing-power growth over long periods.

EXPLORATORY ANALYSIS

Trend comparison dashboard



Gold

Aluminium

Uranium

Silver

Nickel

Long-term winner

Gold ↑

Steady upward trajectory suggests strong hedge characteristics.

Most volatile

Uranium ⚡

Sharp mid-2000s spike highlights speculative energy-cycle behavior.

Industrial sensitivity

Aluminium & Nickel

Prices move with manufacturing demand, supply shocks, and macro cycles.

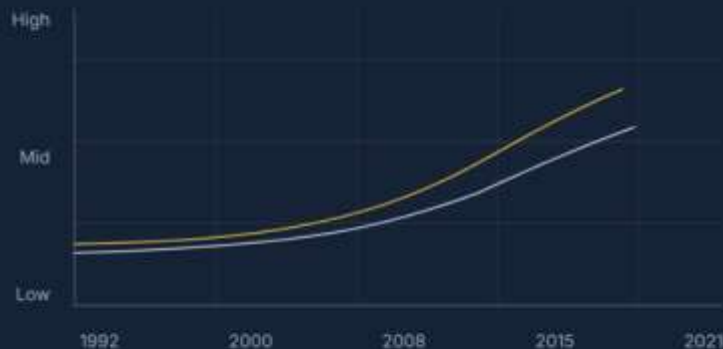
REAL VS NOMINAL

Inflation-adjusted perspective

GOLD

Actual vs inflation-adjusted

Safe-haven lens



Nominal growth looks stronger, but the real-price curve shows that part of the increase reflects the erosion of purchasing power.

URANIUM

Actual vs inflation-adjusted

Speculative cycle



Inflation adjustment softens the apparent peak, helping separate genuine commodity shocks from long-horizon price-level effects.

INSIGHTS

Key market narratives

Gold as a hedge

Gold shows the clearest long-run upward path, making it a useful reference asset during inflation uncertainty and macro stress.

Uranium speculation

Extreme spikes suggest event-driven pricing tied to energy expectations, policy shifts, and sentiment rather than smooth demand growth.

Industrial metal cycles

Aluminium and nickel mirror production cycles, construction momentum, and supply disruptions more than safe-haven behavior.

Why real prices matter

Inflation adjustment prevents overstating long-term performance and supports fairer comparison across decades.

Signal 1

Crisis periods increase volatility

Signal 2

Commodity stories differ by use case

Signal 3

Context beats raw price charts

APPLICATIONS

Business impact and decisions



Investors

Use long-run trend and volatility patterns to evaluate hedge assets, diversify portfolios, and compare commodity cycles.



Manufacturing

Support procurement planning, raw-material budgeting, and timing decisions for cost-sensitive operations.



Supply chain teams

Forecast exposure to price swings and prepare contingency strategies for sourcing and inventory management.



Risk analysis

Identify volatility-prone materials and estimate pressure points during crisis periods and market shocks.



Strategic planning

Translate price behavior into procurement strategy, hedging policy, and scenario-based decision support.



Economic forecasting

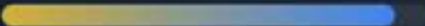
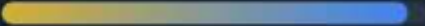
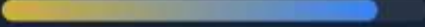
Read commodity movement as a signal of industrial demand, geopolitical tension, and inflation pressure.

What this project developed

Challenges

- **Missing values**
Required interpolation and forward-fill decisions without distorting long-run patterns.
- **Time-series storytelling**
Visuals had to compare multiple metals without overwhelming the audience.
- **Inflation analysis**
Nominal and real value comparisons needed careful explanation for non-technical viewers.
- **Interpretation depth**
Turning price movement into business insight was more valuable than plotting charts alone.

Skills gained

- Data cleaning 
- EDA 
- Visualization 
- Statistical thinking 
- Business storytelling 

Thank You